

SMART CAMPUS COMPLAINT MANAGEMENT SYSTEM

A Desktop Grievance Tracking, Automated Dispatch, and Infrastructure Analytics System

Course:	Java Programming (CSE2001 / CSE2006)
Framework:	VITyarthi 'Build Your Own Project' (BYOP)
Academic Year:	2026 – 2027
Core Technology:	Pure Java (Java SE 17+), Java Swing GUI
Persistence:	Native Java Object Serialization (.dat files)
Architecture:	4-Tier Layered Architecture (UI - Service - Repo - Storage)
Test Suite:	29 Automated Unit & Integration Tests (100% Passing)
Deliverables:	Source Code, Data Store, Design Diagrams, Tests, Report

1. Executive Summary & Introduction

Modern residential universities like VIT Bhopal host thousands of resident students across extensive campus infrastructure. Operational efficiency requires instant detection, tracking, and resolution of infrastructural breakdowns—including electrical hazards, water leakage, Wi-Fi drops, and classroom equipment failures. The Smart Campus Complaint Management System replaces fragmented physical registers with an automated, robust Java desktop system offering end-to-end transparency for both students and campus administration.

2. Problem Statement

Manual complaint handling at university hostels suffers from lost paper registers, delayed escalations, zero student tracking visibility, and a total lack of quantitative performance analytics. Critical safety emergencies (e.g. electrical sparking) compete with routine requests without priority triage. There is an urgent need for a standalone, zero-server-dependency Java system enabling automated ticket generation, controlled state transitions, and real-time maintenance analytics.

3. Project Objectives

- Provide a responsive, professional Java Swing desktop interface for students and campus authorities.
- Automatically generate unique, standardized tracking IDs formatted as CMP-YYYY-XXXX.
- Implement role-based separation of concerns across Student and Administrator personas.
- Enforce valid status transitions (SUBMITTED → IN_PROGRESS → RESOLVED / REJECTED) with administrative remarks.
- Deliver live KPI summaries, resolution rate metrics, and category distribution charts.
- Ensure 100% offline portability and file persistence via safe, atomic Java Object Serialization.

4. Functional Requirements (4 Core Modules)

Module	Key Functional Capabilities
Module 1: Student Management	Student self-registration with RegEx validation (email, 10-digit phone, VIT reg number), credential login, profile inspection (hostel room, department), and session logout.
Module 2: Complaint Management	Ticket filing across 7 categories (Hostel, Electricity, Water, Classroom, Wi-Fi, Cleanliness, Other), 4 priorities (Low, Medium, High, Urgent), atomic ID generation, keyword search, and detailed inspection modal.
Module 3: Admin Management	System-wide complaint directory, departmental work order assignment (moves status to IN_PROGRESS), transition validation, official remarks, 1-click resolution, and linked student info lookup.
Module 4: Reports & Analytics	KPI summary cards (Total, Pending, In-Progress, Resolved, Urgent), Resolution Rate % computation, visual category distribution progress meters, and priority charts.

5. Non-Functional Requirements

- **Usability:** Tailored UI design tokens (Navy, Slate, Emerald, Amber) with styled cards, table badges, and responsive layouts.
- **Reliability:** Atomic file write pattern with temporary files and atomic moves, preventing data corruption upon unexpected termination.
- **Maintainability:** Strict 4-tier layered architecture separating presentation from business logic and data access.
- **Performance:** In-memory LinkedHashMap indexing enabling sub-millisecond table queries and multi-criteria filters.
- **Security & Input Validation:** RegEx validation for email, phone, and registration numbers preventing malformed data.

6. System Architecture & Java OOP Implementation

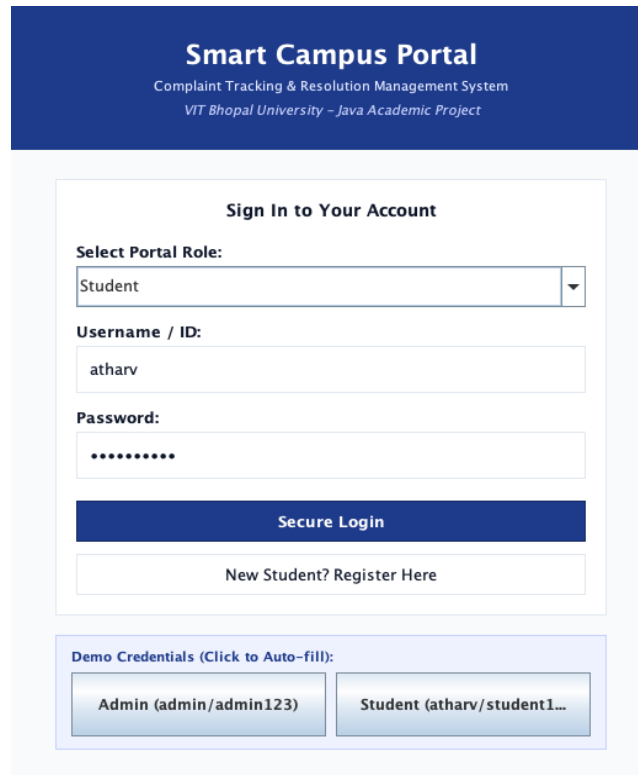
The application strictly enforces a 4-tier layered architecture: Presentation Layer (Swing GUI) → Service Layer (Business Rules & ID Generation) → Repository Layer (Data Access & Synchronization) → File Storage Layer (Binary .dat Files).

Java Concept	Implementation in Codebase
Encapsulation	Private fields with validation guards in setters across User, Student, Admin, Complaint.
Inheritance	Abstract User class extended by concrete Student and Admin subclasses.

Polymorphism	Polymorphic methods <code>getRoleTitle()</code> and <code>getSummary()</code> ; generic Repository CRUD contracts.
Abstraction & Interfaces	Identifiable and Repository decouple high-level services from physical storage.
Method Overloading	6 overloaded search and filter query signatures in <code>ComplaintService.java</code> .
Method Overriding	Overridden <code>toString()</code> , <code>equals()</code> , <code>hashCode()</code> , and <code>compareTo()</code> across domain models.
Collections & Streams	<code>ArrayList</code> , <code>LinkedHashMap</code> , <code>Map</code> , and Java Streams API (<code>Collectors.groupingBy</code> , <code>counting</code>).
Custom Exceptions	Hierarchy rooted in <code>ComplaintManagementException</code> (<code>ValidationException</code> , <code>AuthenticationException</code> , etc.).
Date & Time API	Modern <code>java.time.LocalDateTime</code> and <code>DateTimeFormatter</code> in <code>DateTimeUtil.java</code> .
File Serialization	<code>ObjectInputStream</code> , <code>ObjectOutputStream</code> with atomic move pattern in <code>FileRepository.java</code> .

7. Application Screenshots & Results

Actual screenshots captured directly from the live Java Swing desktop components:



The screenshot displays the 'Smart Campus Portal' login interface. At the top, a dark blue header contains the title 'Smart Campus Portal' and the subtitle 'Complaint Tracking & Resolution Management System VIT Bhopal University - Java Academic Project'. Below this, a white box titled 'Sign In to Your Account' contains a role selection dropdown (set to 'Student'), a 'Username / ID:' field with the value 'atharv', and a 'Password:' field with masked characters. A dark blue 'Secure Login' button is positioned below the password field, followed by a 'New Student? Register Here' link. At the bottom, a 'Demo Credentials (Click to Auto-fill):' section features two buttons: 'Admin (admin/admin123)' and 'Student (atharv/student1...'.

Figure 1: Welcome & Login Portal with Role Switcher and Demo Credentials

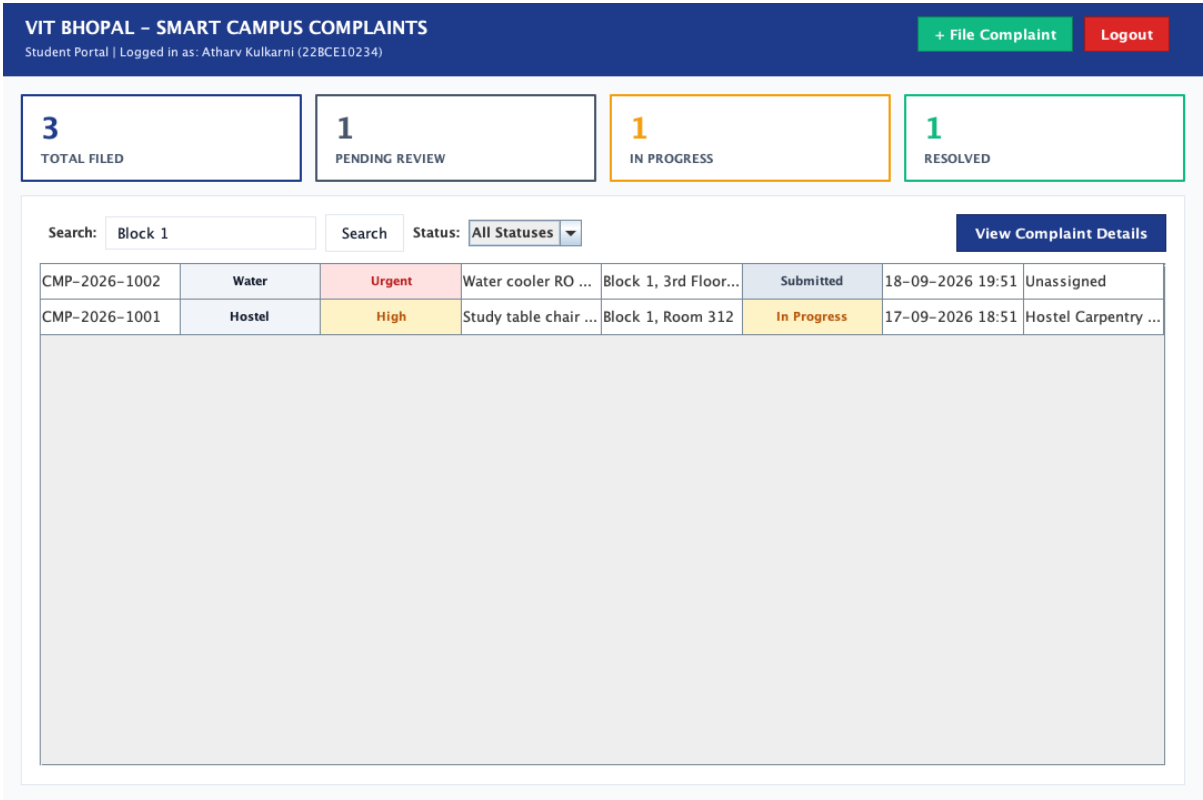


Figure 2: Student Dashboard with KPI Cards and Filterable Complaints Table

VIT BHOPAL – SMART CAMPUS COMPLAINTS

File a New Campus Complaint Ticket

Provide clear details to ensure prompt assignment and resolution by campus maintenance.

Complaint Category *

Water

Severity / Priority Level *

Urgent

Complaint Title (Brief & Clear) *

RO water dispenser leakage causing slippery floor on 3rd floor

Exact Location / Academic Block / Room No *

Boys Hostel Block 1, 3rd Floor Water Station

Detailed Description of Issue *

Continuous water leakage from the bottom pipe fitting of the water cooler dispenser. Water is spilling across the tiles and heading toward staircase #2.

Clear Form

Submit Complaint Ticket

Figure 3: Complaint Submission Form with Category and Priority Tiers

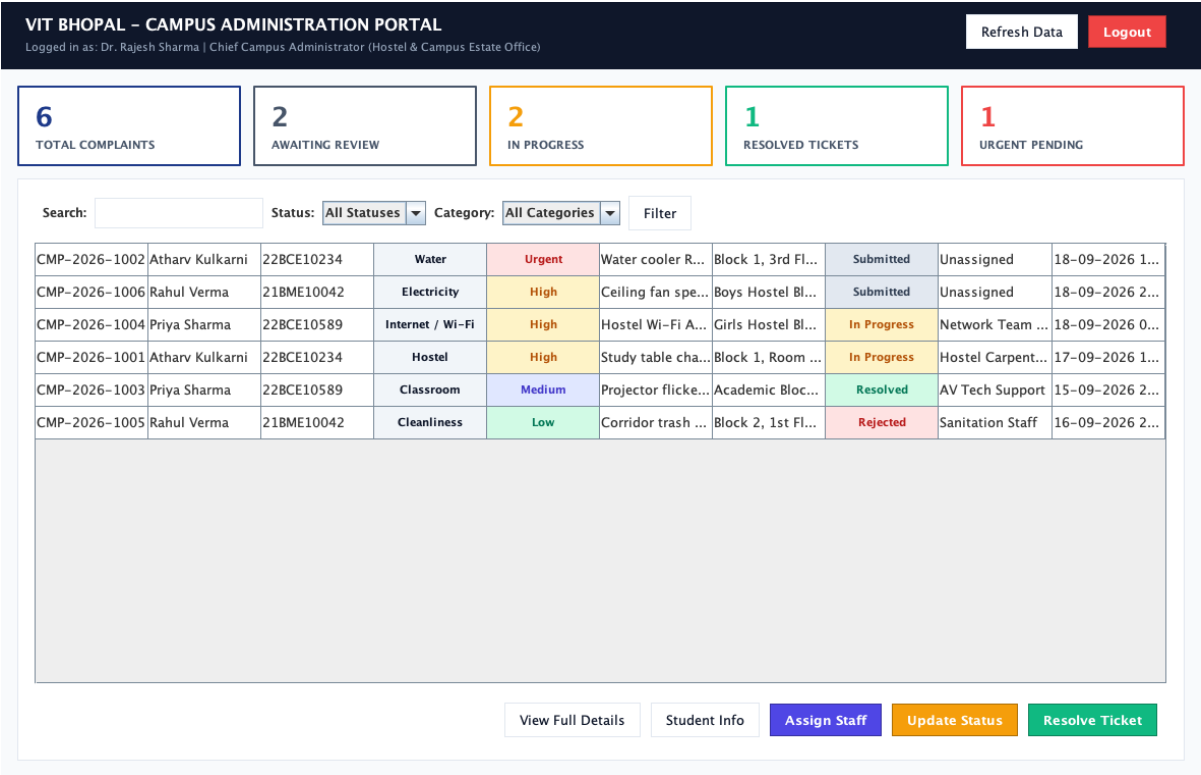


Figure 4: Administrator Management Console with Multi-Criteria Filters & Actions

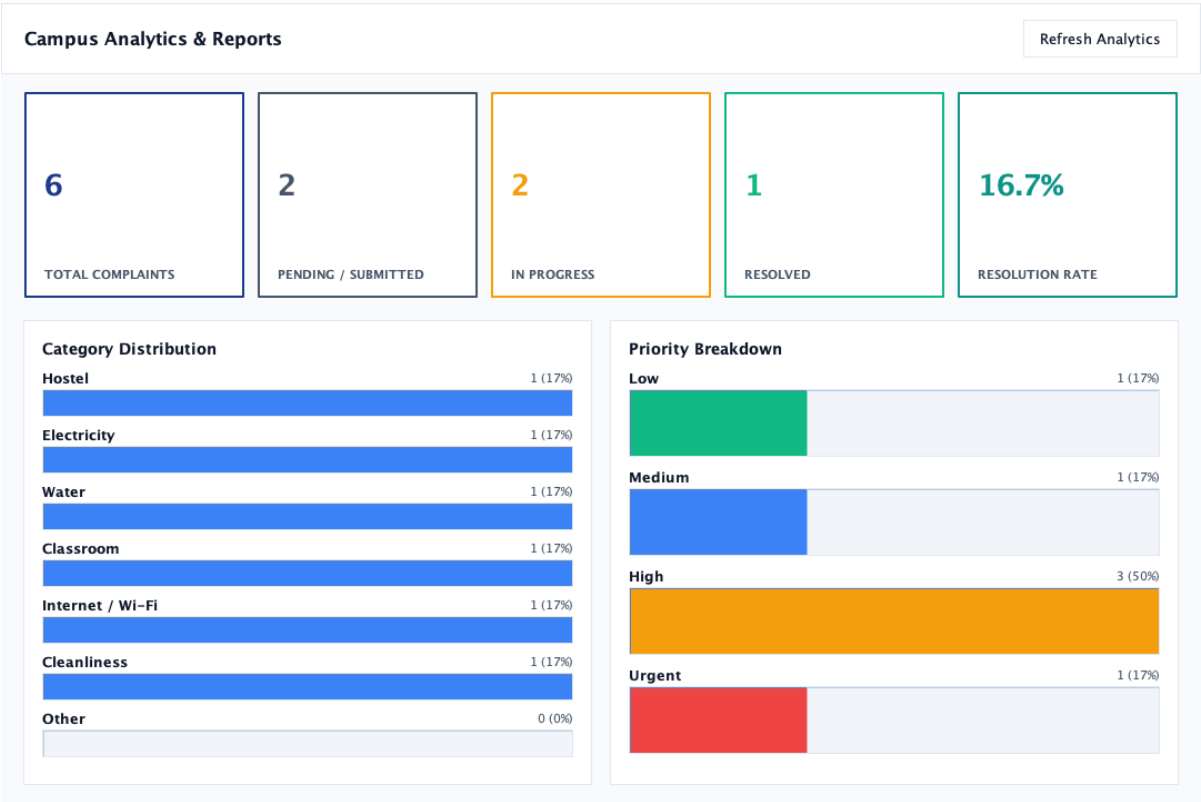


Figure 5: Real-Time Campus Analytics with Category Meters and Resolution Rates

8. Testing Approach & Automated Test Results

The project includes a standalone automated unit and integration test runner (TestSuiteRunner.java) executing 29 tests across all core modules with a 100% pass rate:

#	Test Case Description	Expected Result	Status
1-8	InputValidator: Email, Phone, RegNo, Length boundaries	Valid inputs pass; invalid throw ValidationException	PASS
9-15	AuthService: Student registration, duplicates, login, logout	Accounts created; duplicates prevented; sessions isolated	PASS
16-23	ComplaintService: IDs, transitions, assignments, search	Sequential CMP-YYYY-XXXX generated; transitions guarded	PASS
24-26	FileRepository: Persistence, reload, modification, deletion	Records reload cleanly across fresh repo instances	PASS
27-28	ReportService: Counts, resolution rate %, distribution maps	Accurate formula calculation and 100% key completeness	PASS
29	Integration: End-to-end filing → assignment → resolution → disk reload	Complete lifecycle verified across storage reload	PASS

9. Challenges Faced & Engineering Solutions

- **File Integrity under Sudden Termination:** Directly serializing to live files risks corruption. Solved via atomic write-and-rename pattern utilizing temporary files and Files.move with StandardCopyOption.ATOMIC_MOVE.
- **Thread-Safe Complaint ID Incrementation:** Generated sequential IDs (CMP-YYYY-XXXX) across restarts by initializing an AtomicInteger from the maximum existing ID on storage startup.
- **Modern Swing Aesthetics without Heavy Dependencies:** Default Swing components look dated. Developed a dedicated UITheme token class with custom card borders, table badge renderers, and modern color tokens.

10. Future Enhancements & References

- **Automated SMS/Email Alerts:** Notification triggers when status advances to IN_PROGRESS or RESOLVED.
- **Photo Attachments:** Enabling students to attach image evidence to maintenance complaints.
- **References:** Bloch, Joshua. Effective Java (3rd Ed.); Oracle Java SE 17 Documentation; VITYarthi BYOP Guidelines.